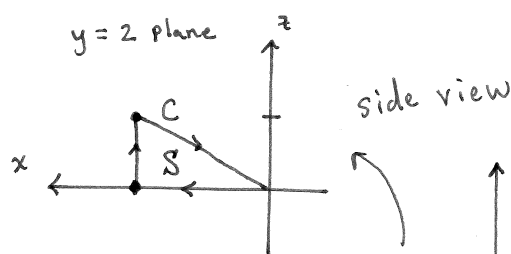


# HW 11 Stokes' Thm

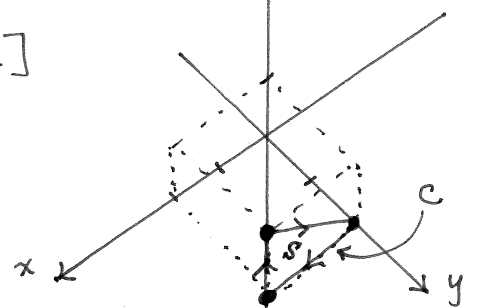


Problem 1:  $\vec{F} = [2y, 3z, x]$

$$(a) \quad \text{curl}(\vec{F}) = [0-3, 0-1, 0-2] = [-3, -1, -2]$$

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S \text{curl}(\vec{F}) \cdot \vec{n} \, dA$$

$$= \iint_S 1 \, dA = \text{Area}(S) = \frac{1}{2}(2)(1) = 1$$



with this orientation, The Normal must point toward the x-axis

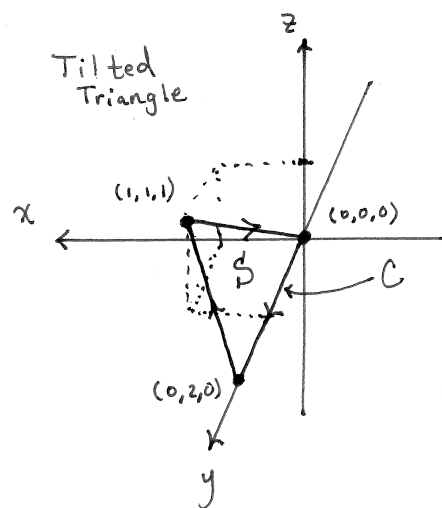
$$\Rightarrow \vec{n} = [0, -1, 0]$$

(unit normal)

$$(b) \quad \text{curl}(\vec{F}) = [-3, -1, -2]$$

$$\text{Note: Area}(S) = \frac{1}{2} \|[2, 0, -2]\|$$

$$= \frac{1}{2} \sqrt{8} = \sqrt{2}$$



with this orientation, the normal should point toward the xy-plane

To find a normal to the triangular plane we compute a cross-product:

$$[0, 2, 0] \times [1, 1, 1]$$

$$= [2, 0, -2]$$

$$\Rightarrow \vec{n} = \left[ \frac{\sqrt{2}}{2}, 0, -\frac{\sqrt{2}}{2} \right]$$

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S \text{curl}(\vec{F}) \cdot \vec{n} \, dA$$

$$= \iint_S -\frac{3\sqrt{2}}{2} + \frac{2\sqrt{2}}{2} \, dA$$

$$= \iint_S -\frac{\sqrt{2}}{2} \, dA$$

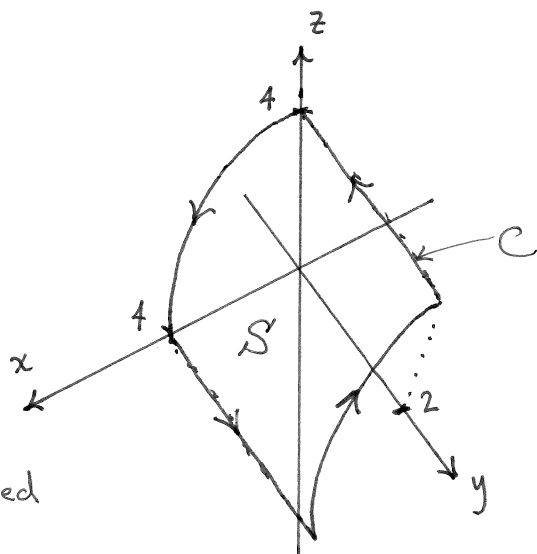
$$= -\frac{\sqrt{2}}{2} \text{Area}(S) = -\frac{\sqrt{2}}{2}(\sqrt{2})$$

$$= -1$$

$$(c) \quad \vec{F} = [y, 2-3yz, x^2+y^2]$$

$$\begin{aligned} \text{curl}(\vec{F}) &= [2y+3y, 0-2x, 0-1] \\ &= [5y, -2x, -1] \end{aligned}$$

Since the unit normal  $\vec{n}$  and  $\text{curl}(\vec{F})$  are not constant, we will need to parameterize  $S$  and compute



$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S \text{curl}(\vec{F})(\vec{r}) \cdot \vec{N} \, du \, dv = (*)$$

Since  $x$  &  $z$  have a circular relationship, consider cylindrical coord.  
Note:  $r=4$  @ every pt of  $S$

$$\begin{array}{lll} x = 4 \cos \theta & u = y & x = 4 \cos v \\ y = y & \longrightarrow & y = u \\ z = 4 \sin \theta & v = \theta & z = 4 \sin v \end{array} \quad \rightarrow \quad \vec{r}(u,v) = [4 \cos v, u, 4 \sin v]$$

$\underbrace{0 \leq u \leq 2 \quad 0 \leq v \leq \frac{\pi}{2}}_{\text{this defines } R}$

$$\begin{aligned} \vec{r}_u &= [0, 1, 0] \\ \vec{r}_v &= [-4 \sin v, 0, 4 \cos v] \end{aligned} \quad \Rightarrow \quad \vec{N} = [4 \cos v, 0, 4 \sin v]$$

A quick check ( $v=0$ ) shows  $\vec{N}$  is pointing in the correct direction.

$$\text{curl} \vec{F}(\vec{r}) = [5u, -8 \cos v, -1]$$

$$\text{curl} \vec{F}(\vec{r}) \cdot \vec{N} = 20u \cos v - 4 \sin v$$

$$\begin{aligned} (*) &= \iint_R 20u \cos v - 4 \sin v \, du \, dv = \int_0^{\frac{\pi}{2}} \int_0^2 20u \cos v - 4 \sin v \, du \, dv \\ &= \int_0^{\frac{\pi}{2}} 10u^2 \cos v - 4u \sin v \Big|_0^2 \, dv \\ &= \int_0^{\frac{\pi}{2}} 40 \cos v - 8 \sin v \, dv \\ &= 40 \sin v + 8 \cos v \Big|_0^{\frac{\pi}{2}} \\ &= (40+0) - (0+8) = 32 \end{aligned}$$

## Problem 2

(a) Show  $\oint_C (f \nabla f) \cdot d\vec{r} = 0$

⊛ Note: Mixed partials are equal since  $f$  has continuous 2<sup>nd</sup> partial derivatives

Consider

$$\begin{aligned} \text{curl}(f \nabla f) &= \text{curl}([ff_x, ff_y, ff_z]) \\ &= [(ff_z)_y - (ff_y)_z, (ff_x)_z - (ff_z)_x, (ff_y)_x - (ff_x)_y] \\ &= [f_y f_z + f f_{zy} - f_z f_y - f f_{yz}, f_z f_x + f f_{xz} - f_x f_z - f f_{zx}, \\ &\quad f_x f_y + f f_{yx} - f_y f_x - f f_{xy}] \end{aligned}$$

⊛

$$= [0, 0, 0]$$

by Stokes' Thm:

$$\oint_C (f \nabla f) \cdot d\vec{r} = \iint_S \text{curl}(\vec{F}) \cdot \vec{n} dA = 0$$

(b) Show  $\oint_C f \nabla g + g \nabla f$

$$\text{curl}(f \nabla g + g \nabla f) = \text{curl}(\nabla(fg)) = 0$$

curl of any gradient is 0